Chaithanya Naik

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Google Scholar

Research Interests

Quantum Computing, Deep Learning, Reinforcement Learning, Scalable systems and Automated Reasoning

Education

University of Wisconsin-Madison 2022-present
PhD student in Computer Sciences
Advisor: Prof. Swamit Tannu

Indian Institute of technology Bombay 2016-2021
M.Tech & B.Tech in Computer Science and Engineering
Minor in Physics
Minor in AI and Data Science
Advisors: Prof. Sai Vinjanampathy, Prof. Ashutosh Gupta , Prof. Amit Sethi

Publications

Scaling Qubit Readout with Hardware Efficient Machine Learning Architectures.
Satvik Maurya, **Chaithanya Naik Mude**, William D. Oliver, Benjamin Lienhard, Swamit Tannu.
*50th International Symposium on Computer Architecture (**ISCA**), 2023, Orlando, Florida, USA.*

Research Experience

Robust Quantum Optimal Control *Spring 2021*
Guide: Prof. Sai Vinjanampathy *Master's Thesis Project, IIT Bombay*
Part of work presented at APS March meeting 2022

- Designed encoder-decoder based Seq2Seq learning model to generate noise-resistent optimized control sequences
- Analyzed literature on Geodesic analysis, Deep Sequential models, and Optimal Quantum Control methods

Controller Synthesis *Spring 2021*
Guide: Prof. Ashutosh Gupta *R&D Project, IIT Bombay*

- Synthesised a controller for real-time system, based on data-driven RL and algorithmic SMT/SAT approaches
- Studied about determinization and minimization of timed automata for a specification given in duration calculus

Interactive Image Segmentation *Fall 2020*
Guide: Prof. Amit Sethi *Bachelor's Thesis Project, IIT Bombay*

- Automated the generation of markers required to segment desired object in image with minimal user intervention
- Modified SeedNet, a DQN approach over MSRA10k dataset to achieve IOU of around 60.5

Professional Experience

VMware Inc. *July 2021 - August 2022*
Avi NSX-T load-balancer | Mentor: Anurag Palsule *Bangalore, India*

- Enhanced the performance of load balancer to manage heavy workloads through kubernetes operator
- Worked on a open source project for a kubernetes operator, catering to needs for NSX-T and openshift platforms
- Worked on functional test-suites to ensure better performance for individual components

Amazon Development Centre India Pvt. Ltd *May - July 2019*
Auto Trouble Ticket Manager | Mentor: Archit Agrawal *Hyderabad, India*

- Automated process of resolving **trouble tickets** raised on **amazon services** by developing java package
- Designed **runtime compilation** module using **Java Compiler API** to implement dynamic code execution
- Developed a **generic framework** to plug-in their models for information retrieval and auto-suggesting resolution

Academic Projects

- **UnrealSynthesisEngine** | Proof Verifier for Unrealizability Logic | Program Synthesis *Fall 2022*
- **Gomoku RL Playing Agent** | Fundamentals of Intelligent agents *Fall 2020*
- **Depth Map Prediction From Single Image** | Computer Vision *Spring 2019*
- **Compiler for C-like language** | Implementation of Programming Languages *Spring 2019*
- **Fake News Detection by Crowdsourcing** | Database and Information Systems *Fall 2018*
- **Micro-architectural Attacks** | FLUSH+RELOAD, DRAMA Template | Computer Architecture *Fall 2018*
- **3D Modelling and Animation** | Computer Graphics *Fall 2018*

Awards & Achievements

- Secured **2nd** position in HCL HACK IITK, the country's largest hackathon in **cybersecurity**, by IIT Kanpur *2022*
- Honourable Mention in **NSUCrypto-2021**, 8th International Olympiad in Cryptography *2021*
- Awarded **Gold** Medal in **Bosch's Route Optimization** challenge at the 8th Inter IIT Tech Meet *2019*
- Secured All India Rank **340** in IIT-JEE Advanced | All India Rank **90** in JEE Main Paper-II *2016*
- Amongst **Top 300** students in the country qualified for **INPhO** (for two straight years) and **INMO** *2015,14*
- Achieved **KVPY** Fellowship with **All India Rank 173** organized under DST, Government of India *2015*
- Recipient of **NTSE** Scholarship awarded by NCERT under Ministry of HRD, Government of India *2012*

Professional Responsibilities

Teaching Assistant

- Data Science Programming I | Prof. Michael Doescher *Fall 2022*
- Quantum Computing and Information | Prof. Sai Vinjanampathy *Spring 2021*
- Python programming lab | Prof. Sai Vinjanampathy *Fall 2020*

Mentor
Quantum Computing, Computer Vision: Maths and Physics Club *April - June 2020*

- Mentored a group of 4 students for the project, Spoof-Resistant Facial Recognition using Deep Learning
- Mentored a group of 6 students for designing Rubik's Cube Solver using Reinforcement Learning
- Mentored a group of 6 students for creating star hopping guide & creating astronomy website using Django
- Helped a total of 12 students in pursuing their interests in Quantum Computing, Astronomy & Computer Vision

Department Academic Mentor | CSE Department, IIT Bombay *June 2020 - July 2021*
Department General Secretary | CSE Department, IIT Bombay *April 2019 - July 2020*

Technical Skills

Programming & Tools	C/C++, Python, Bash, Go-Lang, Qiskit, Git, pennylane
Deep Learning	PyTorch, TensorFlow, Keras, TensorFlow Quantum

Miscellaneous

- Part of the Inter IIT contingent securing **Runner's Up Position** at Inter IIT Tech Meet held at IIT Bombay *2018*
- Attended **Vijyoshi Science Camp** organized by Indian Institute of Science **(IISc)**, Bengaluru, India *2015*